

Technical Report: VMware Virtual Network Segmentation & Lab Configuration

Home Lab Environment for Cybersecurity Research

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Infrastructure: VMware Workstation Pro

Focus: Network Isolation & Penetration Testing Methodologies

1. Lab Overview & Objective

This report documents the complete setup process of a segmented virtual network lab using VMware Workstation. The goal of this lab is to simulate a realistic corporate network environment where a Kali Linux attacker machine is placed in an external network, and the internal network consists of a Domain Controller, Application Server, and Employee System. A pivoting machine (Metasploitable 2 / Web Server) sits between both networks, acting as a bridge. This type of setup is commonly used in penetration testing labs to practice techniques such as network pivoting, lateral movement, and post-exploitation in a completely isolated and safe virtual environment.

2. Network Architecture Design

2.1 Topology Diagram Summary

The overall network is divided into two segments:

- 192.168.50.0/24 (vLan1): External Network containing the Kali Linux attacker machine.
- 10.10.10.0/24 (vLan2): Internal Network containing the Domain Controller, App Server, and Employee System.
- Web Server (Metasploitable 2): Acts as the pivot point with one interface in each network.

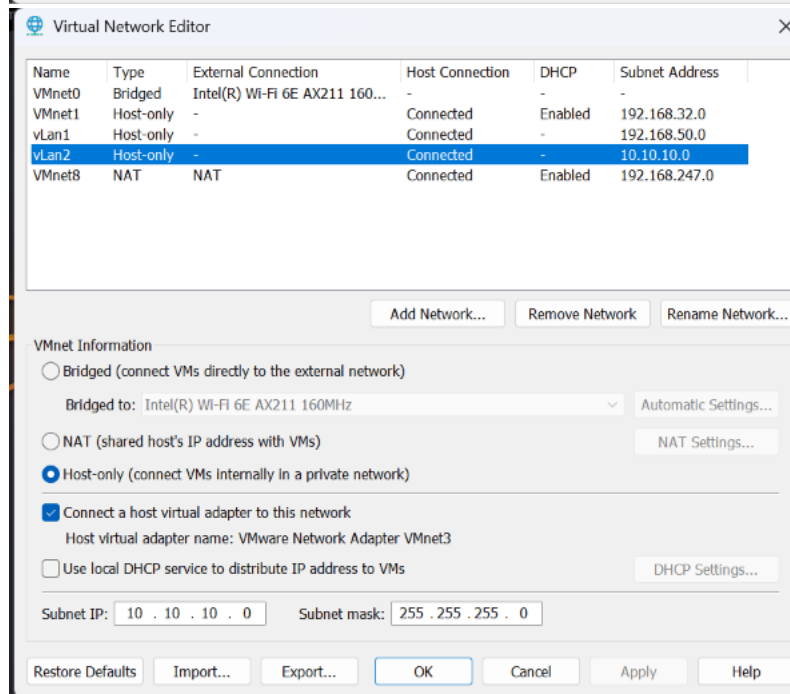
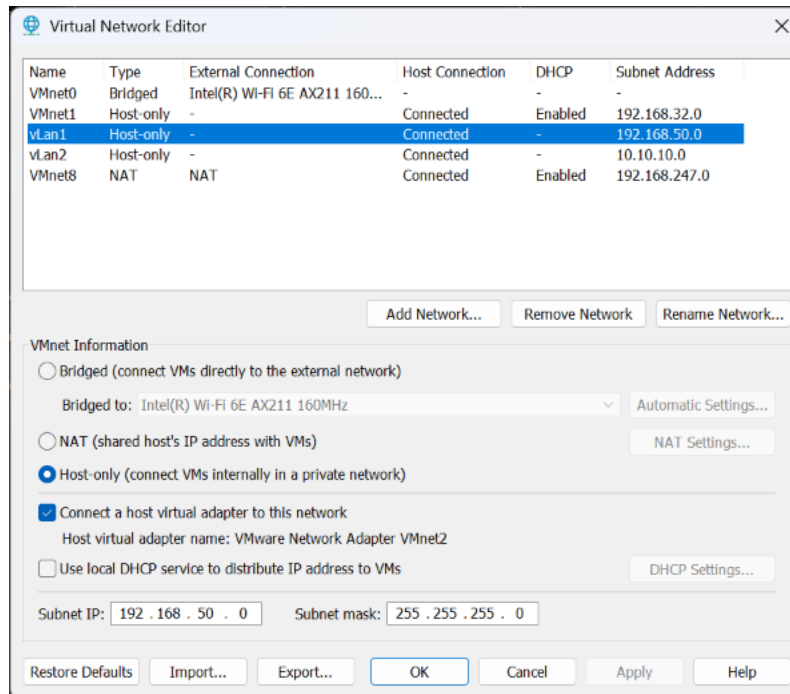
2.2 IP Address Assignment Table

Machine	Network	IP Address	Role
Kali Linux	vLan1 (External)	192.168.50.2	Attacker Machine
Metasploitable 2	vLan1 (External)	192.168.50.3	Web Server / Pivot (eth0)
Metasploitable 2	vLan2 (Internal)	10.10.10.5	Web Server / Pivot (eth1)
Domain Controller	vLan2 (Internal)	10.10.10.2	Windows AD DC
App Server	vLan2 (Internal)	10.10.10.3	Windows Server
Employee System	vLan2 (Internal)	10.10.10.4	Windows 10 Client

3. Step-by-Step Setup Process

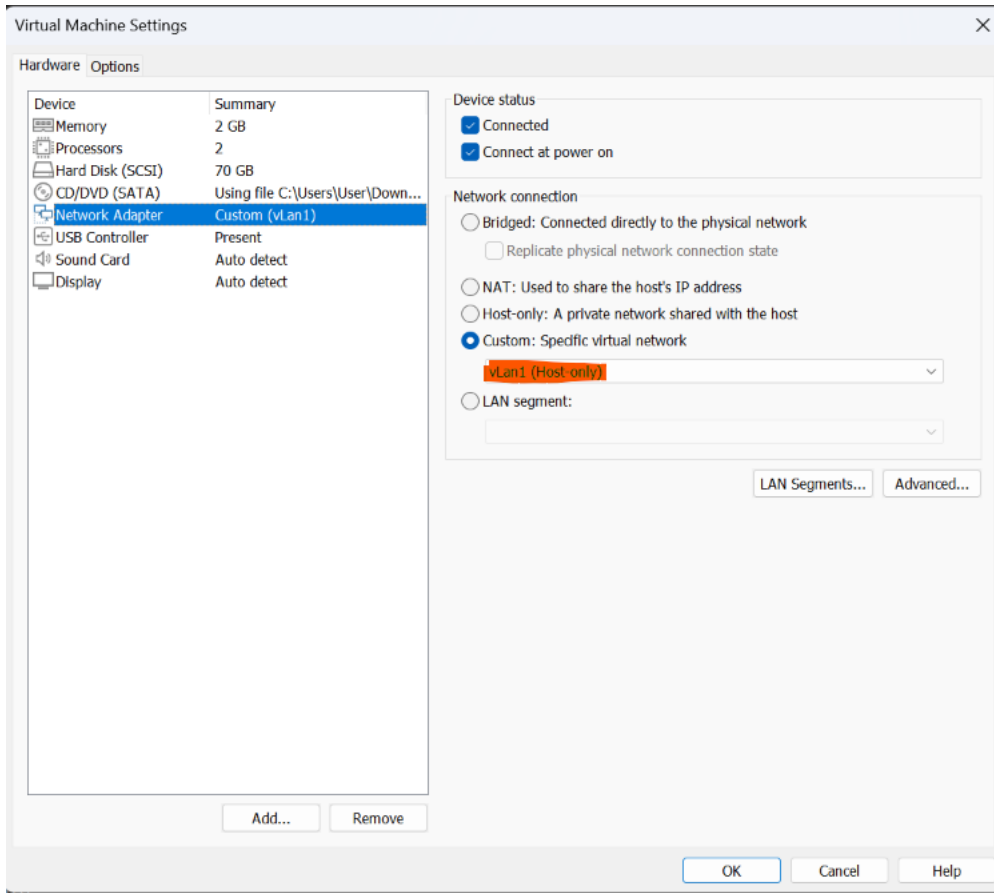
Step 1: Creating Virtual Networks

Opened VMware Virtual Network Editor and created two custom networks (vLan1: 192.168.50.0, vLan2: 10.10.10.0). Both set to Custom with DHCP disabled for manual static IP control.



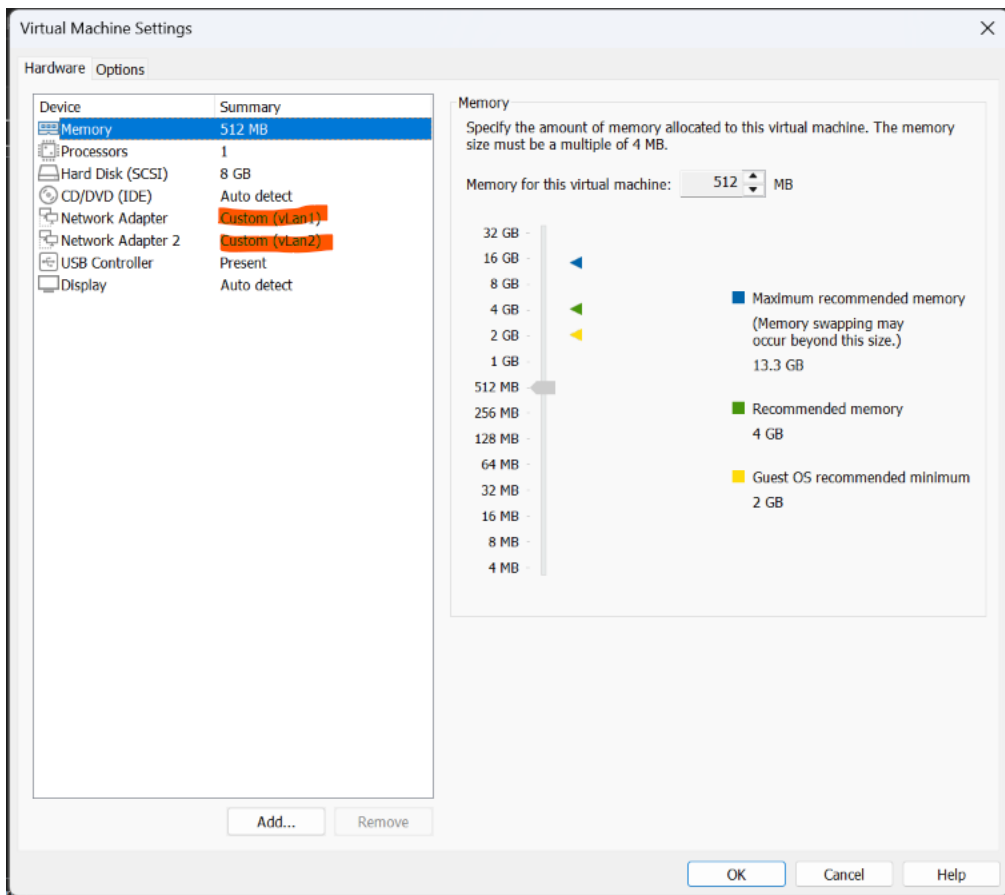
Step 2: Configuring Kali Linux Network

Network adapter configured to Custom (vLan1). Ensures Kali can only reach the external interface of the pivot machine.



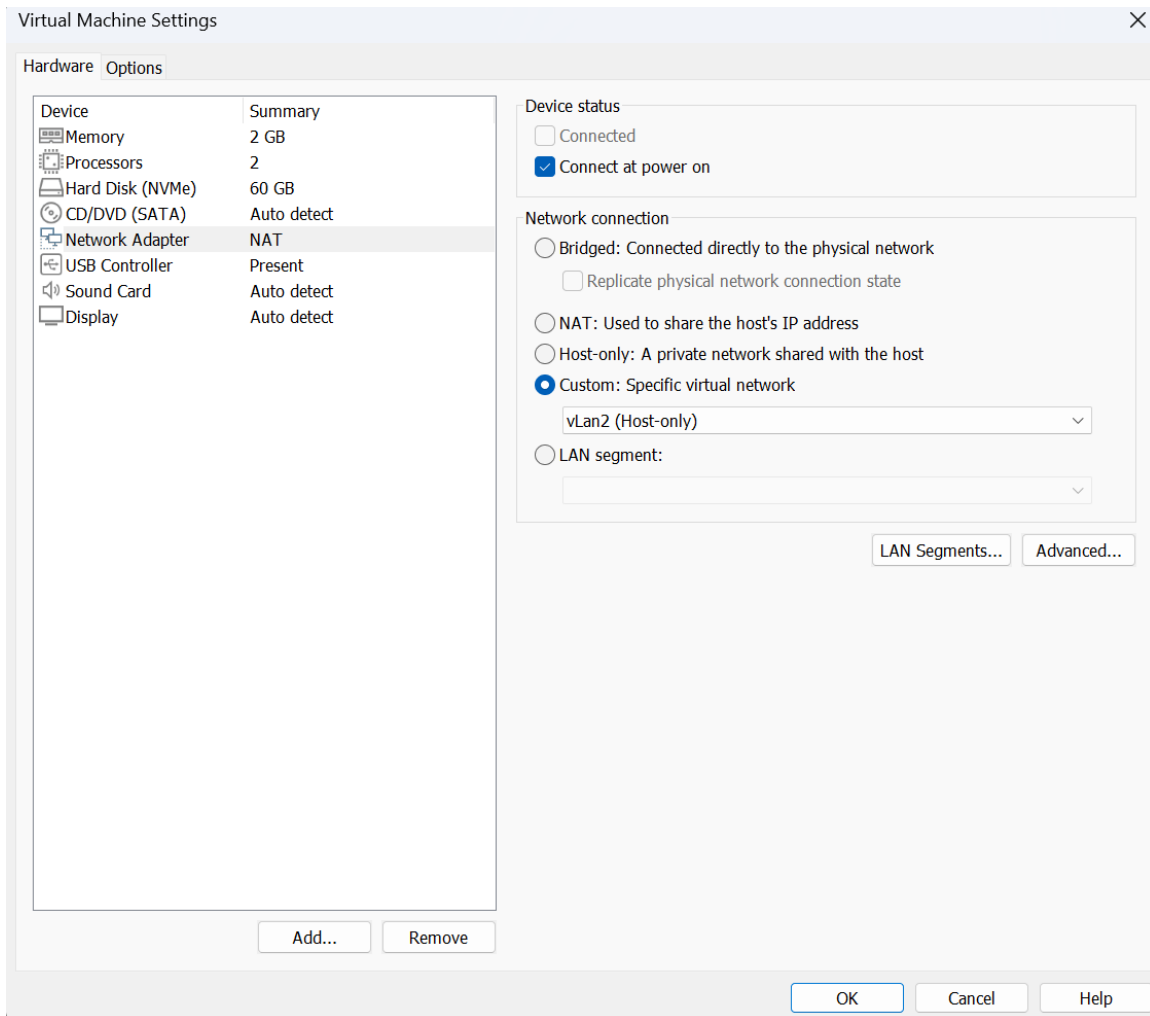
Step 3: Configuring Metasploitable 2 (Pivot)

Configured with TWO network adapters: eth0 on vLan1 (External) and eth1 on vLan2 (Internal). This dual-NIC setup enables bridging between segments.



Step 4: Configuring Windows Machine

Network adapter set to Custom (vLAN2). No direct connection to the external network, simulating an internal-only asset.



Step 5: Static IP on Kali Linux

Edited `/etc/network/interfaces` to assign 192.168.50.2. Verified using `ifconfig`.

```
Session Actions Edit View Help
GNU nano 9.0 /etc/network/interfaces
# This file describes the network interfaces available on your system
# and how to activate them. For more information, see interfaces(5).

source /etc/network/interfaces.d/*

# The loopback network interface
auto lo
iface lo inet loopback

auto eth0
iface eth0 inet static
    address 192.168.50.2
    netmask 255.255.255.0
    gateway 192.168.50.1
#auto eth0
#iface eth0 inet dhcp
```

Step 6: Static IP on Metasploitable 2

Edited /etc/network/interfaces for both eth0 (192.168.50.3) and eth1 (10.10.10.5).
Restarted networking service.

```
GNU nano 2.0.7 File: /etc/network/interfaces
# This file describes the network interfaces available on your system
# and how to activate them. For more information, see interfaces(5).

# The loopback network interface
auto lo
iface lo inet loopback

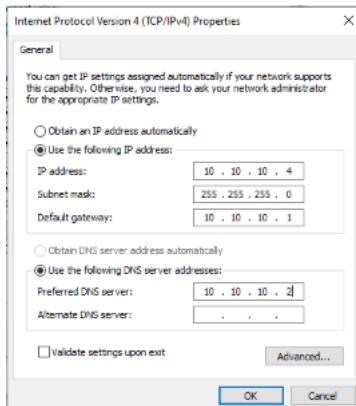
# The primary network interface
auto eth0
iface eth0 inet static
address 192.168.50.3
netmask 255.255.255.0
gateway 192.168.50.1

auto eth1
iface eth1 inet static
address 10.10.10.5
netmask 255.255.255.0
gateway 10.10.10.1

[ Read 19 lines ]
G Get Help  W WriteOut  R Read File  V Prev Page  K Cut Text  C Cur Pos
X Exit      J Justify    W Where Is  O Next Page  U UnCut Text  T To Spell
```

Step 7: Static IP on Windows (Employee System)

Configured via TCP/IPv4 properties (10.10.10.4). DNS set to 10.10.10.2 (Domain Controller).



4. Network Connectivity Logic

From	To	Reachable?
Kali Linux (192.168.50.2)	Web Server eth0 (192.168.50.3)	YES - Same vLan1
Kali Linux (192.168.50.2)	Internal Network (10.10.10.x)	NO - Direct access blocked
Web Server (10.10.10.5)	Internal Network (10.10.10.x)	YES - Same vLan2
Employee System	Web Server eth0 (External)	NO - Different VLAN
Kali via Pivot	Internal Network	YES - After compromising Web Server

5. Key Concepts Practiced

- Network Segmentation: Isolating attacker and target machines.
- Dual-NIC Configuration: Setting up a pivot machine to bridge isolated networks.
- Static IP Assignment: Manual configuration on Linux and Windows.
- Network Pivoting: Reaching internal systems through a compromised bridge.
- DNS Configuration: Proper Active Directory integration for internal name resolution.

6. Conclusion

The lab setup was completed successfully. The architecture mirrors a real-world scenario where an external attacker can reach an internet-facing server but requires pivoting to access internal assets. This serves as an excellent foundation for practicing enumeration, exploitation, and lateral movement.